

EMCH 367 Controls

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Exam #2

Problem 1. (20pt) True or False

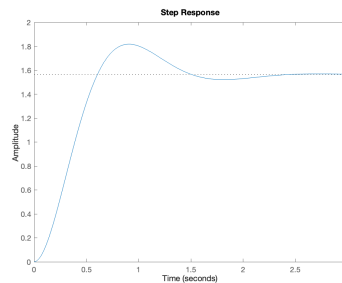
1. For step input, DC motor can hold the position.
2. Type 0 system has no pole at $s = 0$.
3. For Type 1 system, the steady state error for step input is zero.
4. For Type 1 system, the steady state error for ramp input is ∞ .
5. If all coefficients of the characteristic equation are positive, then the system is stable.
6. For $G_{cl}(s) = \frac{G(s)}{1+G(s)H(s)}$, root locus can be found with the angle condition $\angle G(s)H(s) = \pm\pi 2n$, where n is an integer, and magnitude condition $|G(s)H(s)| = 1$.
7. In root locus, the locus start from poles and end at zeros.
8. On real axis, you can use angle condition to find the behavior of root locus.
9. PD controller does not have steady state error.
10. Derivative controller makes the controller more responsive (faster response).

Number	T/F	Number	T/F
1	F	6	F
2	T	7	T
3	T	8	T
4	F	9	F
5	F	10	T

Problem 2. (20pt) Based on the following information on step response of the second order system, identify the natural frequency and the damping ratio.

$$t_p = \frac{\pi}{\omega_d}, \quad t_s = \frac{4}{\zeta\omega_n}, \quad x_p = 1 + e^{-\frac{\zeta}{\sqrt{1-\zeta^2}}\pi}.$$

RiseTime: 0.6048
 TransientTime: 2.0190
 SettlingTime: 2.0190
 SettlingMin: 1.5210
 SettlingMax: 1.8171
 Overshoot: 16.2929
 Undershoot: 0
 Peak: 1.8171
 PeakTime: 0.8980



Solution: Note that the performance indicators are given by

$$t_p = 0.8980 = \frac{\pi}{\omega_d},$$

$$t_s = 2.0190 = \frac{4}{\zeta\omega_n},$$

$$x_p = 1.8171 = 1 + e^{-\frac{\zeta}{\sqrt{1-\zeta^2}}\pi}.$$

Therefore, it follows that

$$\omega_d = \frac{\pi}{0.8980} = 3.4984 = \omega_n \sqrt{1 - \zeta^2},$$

$$\zeta\omega_n = \frac{4}{2.0190} = 1.9812.$$

Thus, we have

$$\frac{1.9812}{\zeta} \sqrt{1 - \zeta^2} = 3.4984 \implies (1.9812)^2 (1 - \zeta^2) = \zeta^2 (3.4984)^2$$

$$\implies \zeta^2 = \frac{1.9812^2}{(3.4984)^2 + (1.9812)^2} \implies \zeta = 0.4928$$

$$\implies \omega_n = \frac{1.9812}{\zeta} = 4.02 \text{ rad/s.}$$

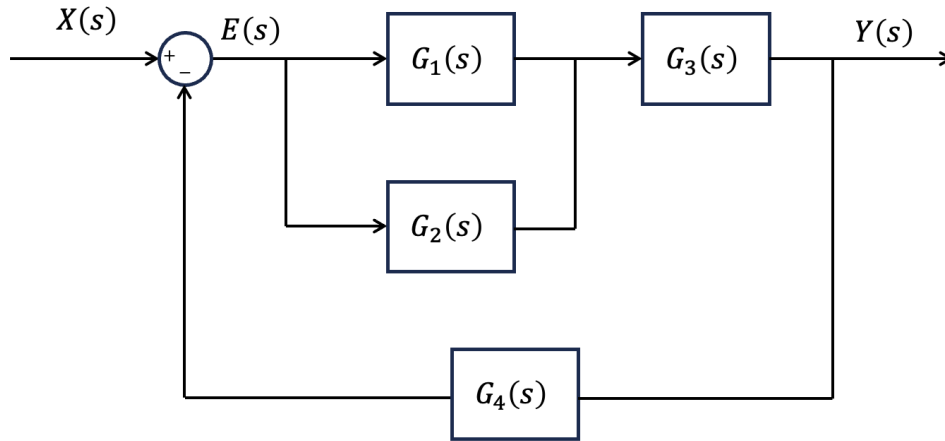
Alternative: If you use the third condition, you will get

$$\zeta = 0.0642, \quad \omega_n = 30.88 \text{ rad/s.}$$

This is due to the error in the simulation.

Problem 3. (30pt)

(a) (15pt) Find $Y(s)/X(s)$.



Solution: Note that

$$\begin{aligned} E_1(s) &= X(s) - G_4(s)Y(s), \\ E_2(s) &= (G_1(s) + G_2(s))E_1(s), \\ Y(s) &= G_3(s)E_2(s) \\ &= G_3(s)(G_1(s) + G_2(s))E_1(s) \\ &= G_3(s)(G_1(s) + G_2(s))(X(s) - G_4(s)Y(s)) \end{aligned}$$

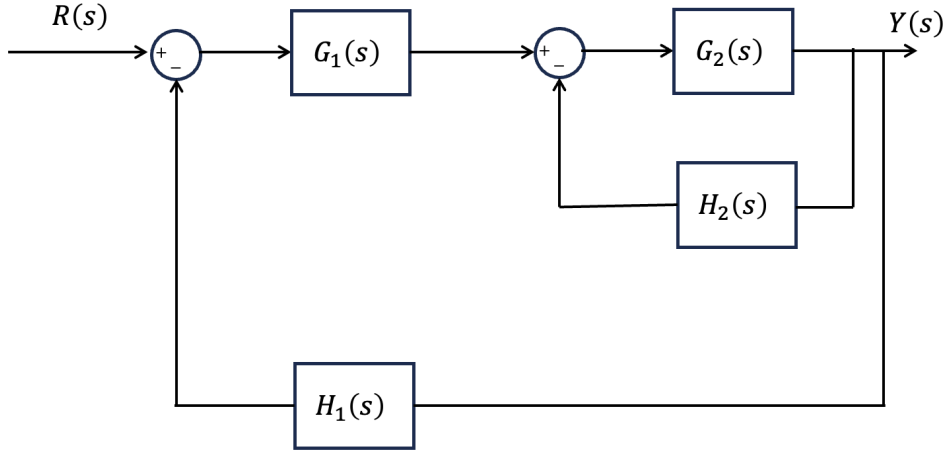
and hence,

$$Y(s)[1 + G_3(s)(G_1(s) + G_2(s))G_4(s)] = G_3(s)(G_1(s) + G_2(s))X(s).$$

Therefore, we have

$$\frac{Y(s)}{X(s)} = \frac{G_3(s)(G_1(s) + G_2(s))}{1 + G_3(s)(G_1(s) + G_2(s))G_4(s)}$$

(b) (15pt) Find $Y(s)/R(s)$.



Solution: Note that

$$E_1(s) = R(s) - H_1(s)Y(s),$$

$$E_2(s) = G_1(s)E_1(s),$$

$$E_3(s) = E_2(s) - H_2(s)Y(s),$$

$$Y(s) = G_2(s)E_3(s)$$

$$= G_2(s)(E_2(s) - H_2(s)Y(s))$$

$$= G_2(s)(G_1(s)E_1(s) - H_2(s)Y(s))$$

$$= G_2(s)(G_1(s)(R(s) - H_1(s)Y(s)) - H_2(s)Y(s))$$

$$= G_2(s)G_1(s)R(s) - G_2(s)G_1(s)H_1(s)Y(s) - G_2(s)H_2(s)Y(s),$$

and hence,

$$Y(s)[1 + G_2(s)G_1(s)H_1(s) + G_2(s)H_2(s)] = G_2(s)G_1(s)R(s).$$

Therefore, we have

$$\frac{Y(s)}{R(s)} = \frac{G_2(s)G_1(s)}{1 + G_2(s)G_1(s)H_1(s) + G_2(s)H_2(s)}$$

Problem 4. (30pt) Consider the following transfer function

$$G(s) = \frac{K \frac{s + 0.5}{s^2(s + 2)(s + 5)}}{1 + K \frac{s + 0.5}{s^2(s + 2)(s + 5)}}$$

(a) (10pt) Determine the root locus on the real axis.

Solution: The characteristic equation is given by

$$1 + K \frac{s + 0.5}{s^2(s + 2)(s + 5)} = 0.$$

Note that the angle condition is given by

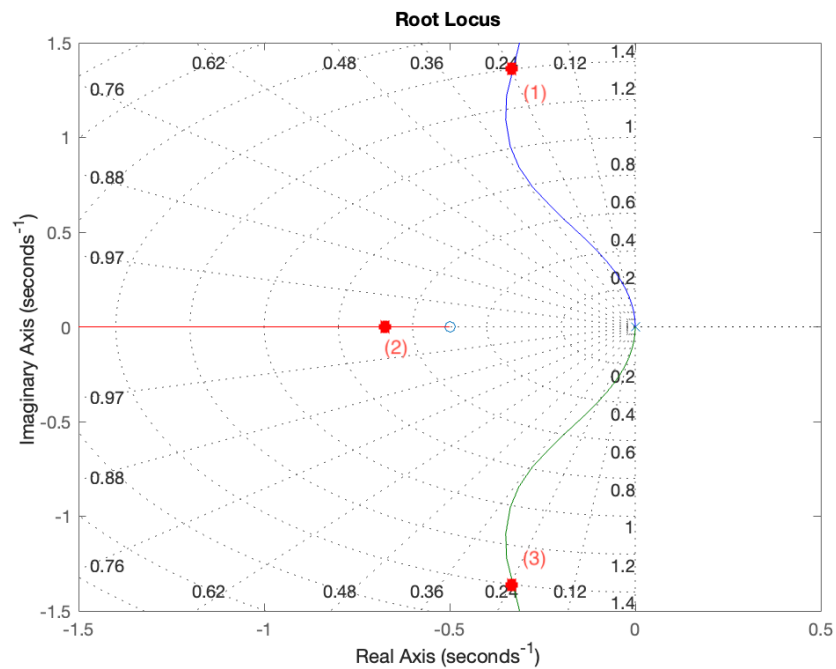
$$\angle(s + 0.5) - 2\angle(s) - \angle(s + 2) - \angle(s + 5).$$

Therefore, on the real axis, we can find that

$$\left\{ \begin{array}{ll} s > 0 : & 0 - 0 - 0 - 0 = 0 \\ -0.5 < s < 0 : & 0 - 2\pi - 0 - 0 = -2\pi \\ -2 < s < -0.5 : & \pi - 2\pi - 0 - 0 = -\pi \\ -5 < s < -2 : & \pi - 2\pi - \pi - 0 = -2\pi \\ s < -5 : & \pi - 2\pi - \pi - \pi = -3\pi. \end{array} \right.$$

Therefore, on the real axis, the angle condition is satisfied for $s < -5$ and $-2 < s < -0.5$.

(b) (10pt) The roots for $K = 15$ is marked with asterisks. Which roots represent the dominant poles? (select all)

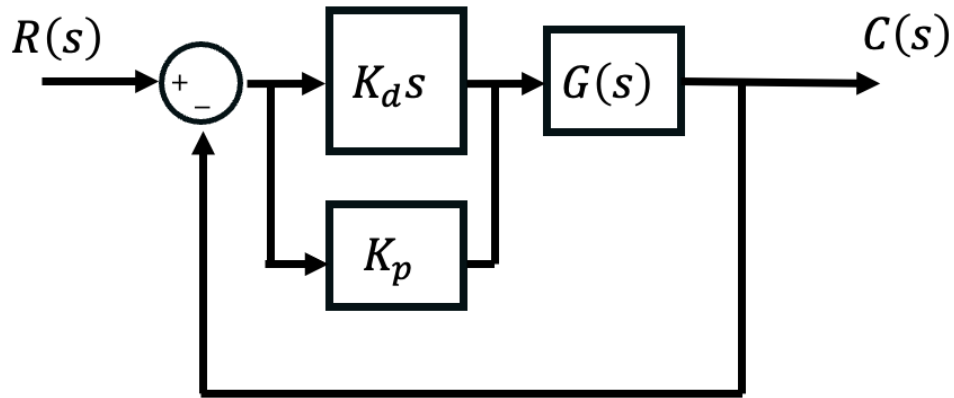


Solution: The dominant poles are (1) and (3).

(c) (10pt) Find the damping ratio of the dominant poles.

Solution: It follows from the figure that the damping ratio is 0.24.

Bonus Problem. (20pt) Consider the following block diagram.



Let the plant be

$$G(s) = \frac{1}{s^2 + 5s}$$

and $K_d = 3$. If we want the closed loop system to have damping ratio of $\zeta = 0.8$, what should be the value of K_p ?

Solution: Note that

$$\begin{aligned} G_{cl}(s) &= \frac{(K_p + K_d s) \frac{1}{s^2 + 5s}}{1 + (K_p + K_d s) \frac{1}{s^2 + 5s}} \\ &= \frac{K_p + K_d s}{s^2 + (5 + K_d)s + K_p} \\ &= \frac{3s + K_p}{s^2 + 8s + K_p}, \end{aligned}$$

and hence, to have $\zeta = 0.8$, note that

$$2\zeta\sqrt{K_p} = 8 \implies K_p = 25.$$